

Anatomy Of A Modular Toolchain For Reproducible Bioinformatics Workflows

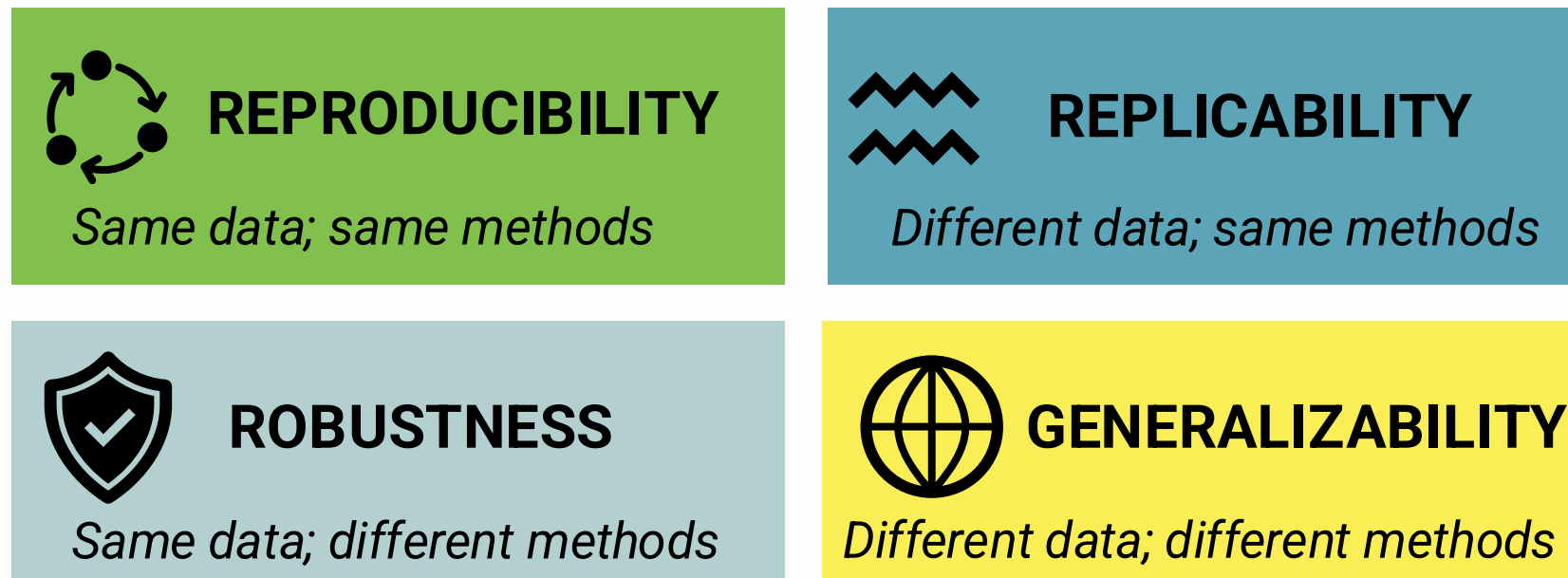
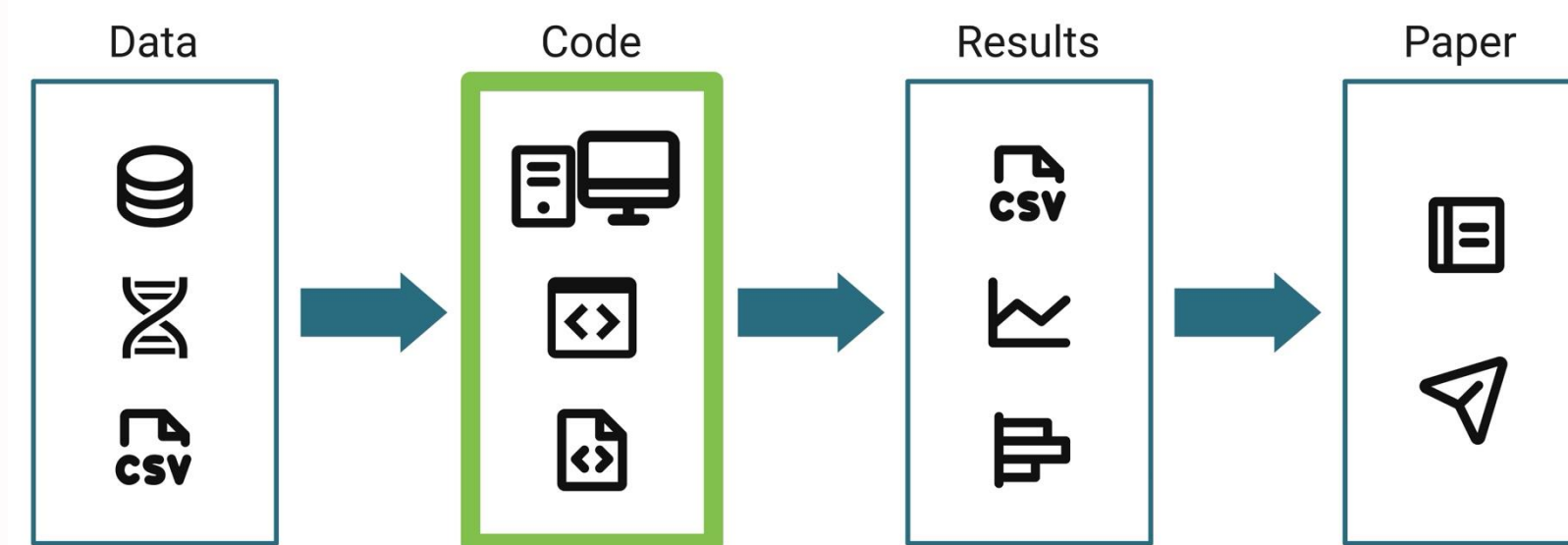
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INTRODUCTION

Reproducibility is the foundation of trustworthy data-intensive biological research. Our framework implements core reproducibility principles across our software suite.



Schloss PD. 2018. 10.1128/mBio.00525-18.

BEST PRACTICES

Version Control

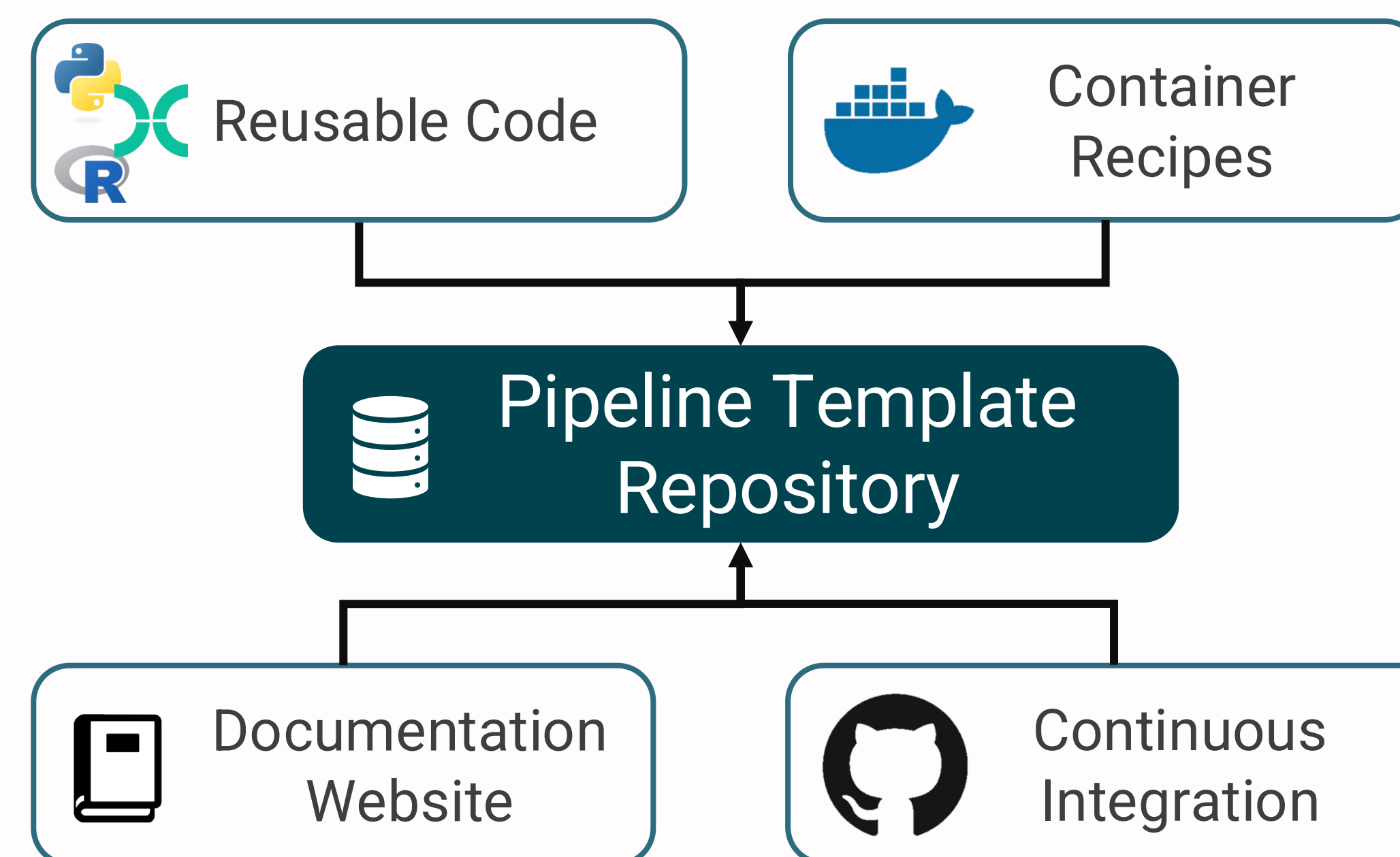
Modular Design

Automated Tests

Documentation

We enforce consistency through templates, modularity, and automation to prevent technical debt, streamline maintenance, and enable long-term sustainability.

TOOLCHAIN FOR REPRODUCIBLE BIOINFORMATICS



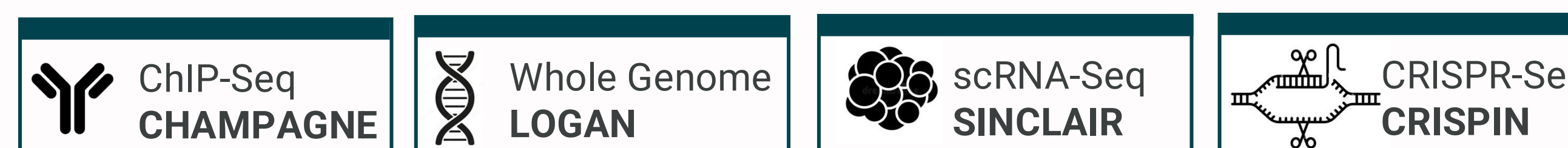
Unified framework from template to deployment with automated quality checks.

Fix once;
deploy everywhere

Consistent CLI &
Implementation

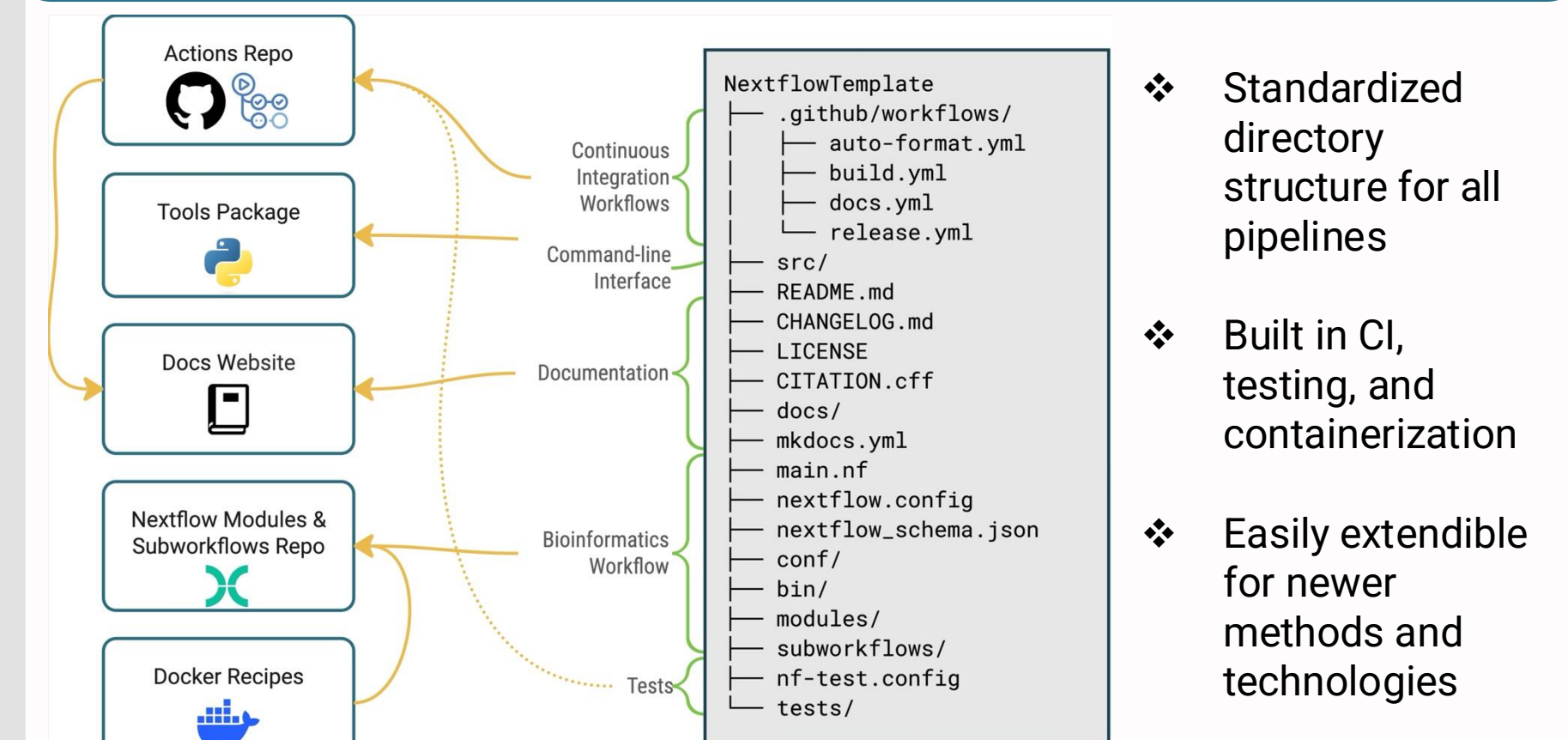
Automated testing &
quality checks

PIPELINES BUILT

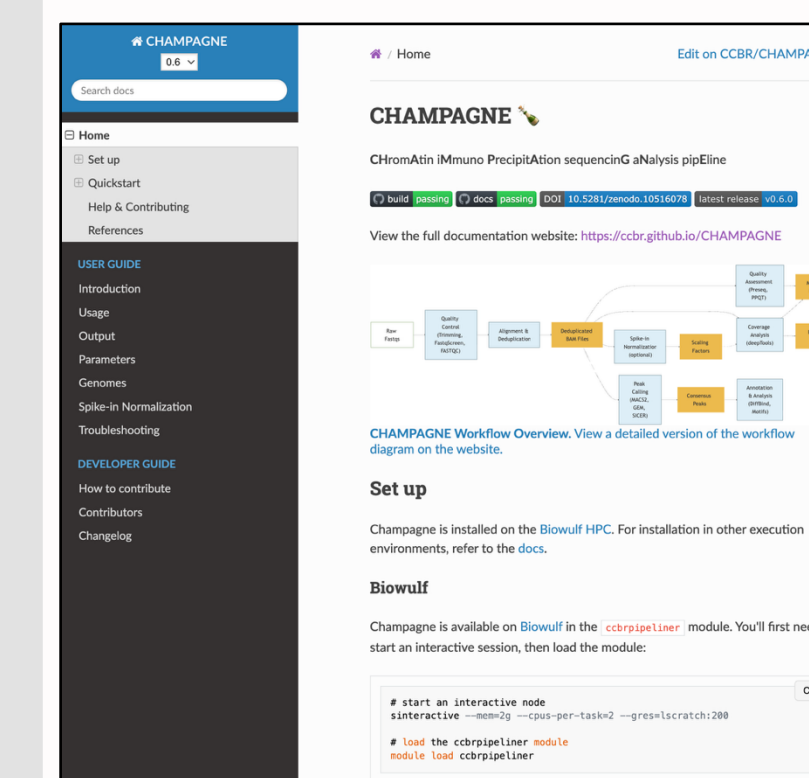


10+ pipelines on NIH Biowulf HPC cluster
\$ module load ccbripipeline

PIPELINE TEMPLATE



CLI, DOCUMENTATION, AUTOMATION



```
$ champagne run --input samplesheet.csv
$ Sinclair run --input samplesheet.csv -params-file ...
```

Consistent usage guidelines and documentation across all pipelines simplifies user training.

GitHub Actions runs CI checks on every change.

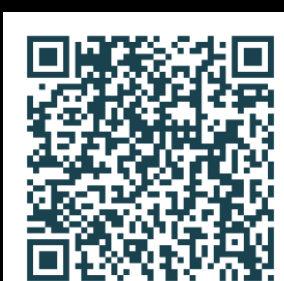
Scalable, Reproducible, Sustainable Bioinformatics

NIH Biowulf HPC
Secure, Scalable, and high performance computing platform

GitHub Open Source
Collaborate, Contribute, Reuse

Documentation
Guides, Tutorials, Best Practices

This Poster



Template Repo



*Advancing reproducible computational research through
high-quality bioinformatics software and data analysis pipelines*

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